Aryan Minhas

613-447-5513 | aryanminhas13@gmail.com | Personal Website
linkedin.com/in/aryanminhas | github.com/aryanminhas13

Education

University of Ottawa Ottawa, ON
B.Sc. Computer Science (Honours, Co-op) – Dean's List, CGPA: 8.13/10 *2023 – 2027*
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Databases, Computer Architecture

Experience

Data Scientist Co-op Sept. 2026 – Present
Statistics Canada *Ottawa, ON*

- Identified over **$10,000+ in annual storage savings** by developing a duplicate-file detection workflow that analyzed enterprise file inventories and surfaced high-value cleanup opportunities.
- Analyzed large administrative datasets in **Python** to identify trends, outliers, and data-quality issues, then summarized findings for downstream analysis and reporting.
- Built a **Python data pipeline** using **pandas** and **PyArrow** to ingest Parquet and tabular datasets, standardize schemas, merge multi-source records, and generate validated analysis-ready outputs for downstream statistical workflows.

Software Engineer / Data Engineer Co-op May 2026 – Aug. 2026
Department of National Defence *Ottawa, ON*

- Executed service-intake data migration with record reconciliation and preservation of approval metadata, lifecycle states, and attachments; achieved **zero data loss** and eliminated **$20,000+ in annual licensing costs**.
- Integrated project and operational datasets into executive reporting dashboards, consolidating delivery KPIs, expenditure, and risk indicators to support portfolio-level decision-making.
- Automated work-item creation from sprint, meeting, and initiative records, translating operational inputs into structured backlog entries for capacity planning and delivery tracking.

Automation Developer Co-op Jan. 2026 – Apr. 2026
Department of National Defence *Ottawa, ON*

- Engineered service-intake and approval workflows with conditional routing, lifecycle-state synchronization, and automated handoffs between business and technical teams.
- Implemented API-driven pull-request review notifications with duplicate-alert suppression, reducing review response time by **80%**.
- Developed an **LLM-powered document-processing pipeline** to parse government filings, extract structured data, validate outputs, and automate downstream routing.

Projects

Distributed Job Orchestration Platform | *Go, React, TypeScript, Redis, PostgreSQL, Docker*

- Built a distributed job-processing platform with a React dashboard and Go REST API supporting scheduling, retries, cancellation, persistent queues, and worker-failure recovery.
- Implemented real-time status, logs, and metrics with WebSockets; load-tested **100,000+ jobs** at **8,000+ jobs/minute** with sub-**150 ms p95 dispatch latency**.
- Containerized scalable workers with Docker and benchmarked eight concurrent workers, achieving 6.2× **throughput scaling** while persisting execution metadata in PostgreSQL.

Packet Panic | *JavaScript, HTML5 Canvas, CSS, localStorage*

- Built a browser arcade game with Canvas rendering, collision detection, progressive difficulty, and keyboard/touch controls; instrumented gameplay with event IDs, session IDs, timestamps, and outcome metadata.
- Implemented a bounded **4,000-event** telemetry store and session-level aggregation to calculate wave-progression funnels, survival duration, interception rates, and event distributions in an interactive analytics dashboard.

Awards & Achievements

CATAN NATIONAL CHAMPION – Canadian national champion; qualified for international competition in Brazil and Scotland through strategic planning and decision-making under uncertainty.

Technical Skills

Languages: Python, Java, Go, TypeScript, JavaScript, SQL, HTML/CSS
Frameworks/Libraries: React, Node.js, Express, Flask, scikit-learn, Pandas, NumPy, PyArrow
Databases/Platforms: PostgreSQL, Redis, Firebase, Azure DevOps, SharePoint, Power Platform, Power BI
Developer Tools: Git, Docker, REST APIs, WebSockets, OData, Postman